OPEN SOURCE — SYSTEM ARCHITECTURE APPENDIX

Chen Xingqiang · diagrams restyle DS4/BW01 · content from each repo's official architecture + source tree

TRACESMITH
docs/architecture.drawio
include/tracesmith/*

YIRAGE YPK
docs/yirage-architecture.drawio
include/backend/*.h

JAX-SKLEARN
xlearn/_jax/*.py
xlearn/__init__.py

SECRET-LEARN
docs/secret_learn_architecture.drawio
src/secretlearn/

YOUTUBE-DNN
docs/architecture.drawio
src/python + src/scala

RESEARCH CLI
packages/cli + packages/core
modelRouterService.ts

QUESTIONS-GEN
src/QuestionsGenTrainer.py
questions_gen/

STYLE
dark canvas · nested containers · Courier · real paths · no invented layers

TRACESMITH — SYSTEM ARCHITECTURE

GPU Profiling & Replay · C++17 core + pybind11 · source: docs/architecture.drawio · github.com/chenxingqiang/TraceSmith

APPLICATION LAYER

C++ Application
user binary

Python Application
import tracesmith

C++ CLI
cli/main.cpp

Python CLI
python/tracesmith/cli.py

External Tools
Perfetto UI / Chrome

PYTHON BINDINGS

pybind11 module
python/src/bindings.cpp

_tracesmith.so
native extension

python/tracesmith/__init__.py
package API

native

TRACESMITH CORE LIBRARY (C++17) include/tracesmith + src/

CUPTIProfiler
src/capture/cupti_profiler.cpp

MCPTIProfiler
src/capture/mcpti_profiler.cpp

MetalProfiler
src/capture/metal_profiler.mm

AscendProfiler
src/capture/ascend_profiler.cpp

ROCmProfiler
src/capture/rocm_profiler.cpp

BPFTracer
src/capture/bpf_tracer.cpp

MemoryProfiler
src/capture/memory_profiler.cpp

events

RingBuffer SPSC
common/ring_buffer.hpp

Types / TraceEvent
src/common/types.cpp

StackCapture
src/common/stack_capture.cpp

XRayImporter
src/common/xray_importer.cpp

SBTWriter/Reader
src/format/sbt_format.cpp

GPUStateMachine
src/state/gpu_state_machine.cpp

replay

InstructionStream
src/state/instruction_stream.cpp

TimelineBuilder
src/state/timeline_builder.cpp

TimelineViewer
src/state/timeline_viewer.cpp

PerfettoExporter
src/state/perfetto_exporter.cpp

PerfettoProto
src/state/perfetto_proto_exporter.cpp

ReplayEngine
src/replay/replay_engine.cpp

StreamScheduler
src/replay/stream_scheduler.cpp

OperationExecutor
src/replay/operation_executor.cpp

DeterminismChecker
src/replay/determinism_checker.cpp

FrameCapture
src/replay/frame_capture.cpp

GPUTopology
src/cluster/gpu_topology.cpp

MultiGPUProfiler
src/cluster/multi_gpu_profiler.cpp

TimeSync NTP/PTP
src/cluster/time_sync.cpp

NCCLTracker
src/cluster/nccl_tracker.cpp

GDB RSP
src/gdb/rsp_handler.cpp

TracyExporter
src/tracy/tracy_exporter.cpp

GPU BACKENDS & SYSTEM (createProfiler + nsys / mcTracer / xctrace / msprof)

NVIDIA CUDA
CUPTI SDK + nsys

Apple Metal
Metal API + xctrace

MetaX MACA
MCPTI + mcTracer

Huawei Ascend
CANN + msprof

AMD ROCm
roctracer

Linux eBPF
kernel events

libunwind
host stacks

Perfetto SDK
protobuf

OUTPUT FORMATS

trace.sbt
SBT binary

trace.json
Perfetto JSON

trace.perfetto
protobuf

graph.dot
Graphviz

ASCII timeline
terminal

comm matrix
NCCL heatmap

v0.10+ · capture→ring→SBT→state/timeline→replay/export · files from include/tracesmith and src/

YIRAGE (YPK) — SYSTEM ARCHITECTURE

RuntimeFusion × Multi-Backend · source: docs/yirage-architecture.drawio + include/backend/*.h · github.com/chenxingqiang/YiRage

INFERENCE ENGINE (scheduling + KV owner)

- **vLLM** — continuous batch · PagedAttention
- **SGLang** — RadixAttention · extend meta
- **meta** — block_table · seq_lens · SM budget

decode

RUNTIMEFUSION yirage.serving · RF.step(inputs, meta) selects 0..N capsules / decode step

- **RF.step()** — yirage.serving
- capsule
- **FusionCapsule MLP** — gate_up + down yirage.core
- **FusionCapsule seg** — gate_up → down pipeline
- **skip / fallback** — engine native ops

plan

SEARCH & PLAN (offline / warm-up)

- **superoptimize(backend)** — KN/TB graph search
- **FusionPlan / μGraph** — profile + fingerprint verify
- **MuGraph cache** — shape-bucket restore
- **Ray coordinator** — distributed search optional
- **serving tiers** — seed_verify · full_tb · full_tb_ray

PYTHON API & BACKEND MANAGER (C++)

- **PersistentKernel** — python/yirage
- **KNGraph / TBGraph** — transpiler → kernels
- **BackendRegistry** — include/backend/backend_registry.h
- **yirage.core** — Cython runtime
- **KernelExecutor factory** — search + execute same backend

BACKEND IMPLEMENTATIONS include/backend/*.h + include/kernel/{cuda,maca,cpu,mps,ascend}/

- **CUDA** — cuda_backend.h warp 32 nvcc
- **MACA** — maca_backend.h warp 64 mxcc
- **ROCm** — rocm_backend.h AMD GPU
- **MPS** — mps_backend.h Apple GPU
- **Ascend** — ascend_backend.h CANN
- **CPU** — cpu_backend.h OpenMP/MKL
- **Triton** — triton_backend.h DSL
- **NKI** — nki_backend.h SBUF/DMA
- **cuDNN** — cudnn_backend.h
- **MKL** — mkl_backend.h
- **TPU** — tpu_backend.h
- **XPU** — xpu_backend.h
- **MLIR** — mlir_backend.h CPU JIT
- **FPGA** — fpga_backend.h experimental

HARDWARE TARGETS

- **NVIDIA GPU** — Ampere / Hopper
- **MetaX C500/C550** — MACA
- **x86 / ARM CPU** — Intel / AMD / Apple
- **Apple Silicon** — M1–M4
- **Huawei Ascend** — 910 / 910B / 310P
- **AWS Neuron / Intel XPU** — NKI

Engine owns scheduling & KV; RF owns fusion identity per step · BackendRegistry search+execute same backend

JAX-SKLEARN (xlearn) — SYSTEM ARCHITECTURE

drop-in sklearn · files: xlearn/__init__.py + xlearn/_jax/*.py · github.com/chenxingqiang/jax-sklearn

USER CODE

import xlearn as sklearn
public API

LinearRegression / KMeans / PCA
xlearn.linear_model · cluster · decomposition

set_config / config_context
xlearn/_jax/_config.py

PACKAGE INIT xlearn/__init__.py

__getattr__(name)
lazy submodule import

inject

_JAX_ENABLED
import xlearn._jax

_auto_jax_accelerate_module
replace estimator classes

AcceleratorRegistry
xlearn/_jax/_accelerator.py

PROXY xlearn/_jax/_proxy.py

create_intelligent_proxy
wrap estimator type

EstimatorProxy.__init__
_create_implementation()

JAX

JAX path
create_accelerated_estimator

fallback

sklearn fallback
original_class(*args)

JAX KERNELS xlearn/_jax/

_universal_jax.py
generic mixins JIT/vmap

_accelerator.py
dispatch + registry

_data_conversion.py
NumPy ↔ JAX

_gp.py
Gaussian process

_mlp.py
neural_network

_svm.py
SVM

_tree.py
trees

CONFIG / HEURISTICS xlearn/_jax/_config.py

enable_jax
on/off

jax_platform
cpu | gpu | tpu | metal

LinearRegression
min_complexity 1e8

KMeans
min_complexity 1e6

PCA
min_complexity 1e7

HARDWARE (JAX backends)

CPU jaxlib
default

CUDA jax[gpu]
NVIDIA

TPU jax[tpu]
Google Cloud

Metal jax-metal
Apple experimental

Injection is lazy: xlearn.__getattr__ → proxy class in module → fit() chooses JAX or sklearn

SECRET-LEARN — SYSTEM ARCHITECTURE

Privacy-preserving ML · source: docs/secret_learn_architecture.drawio + src/secretlearn/ · github.com/chenxingqiang/Secret-Learn

APPLICATION & USE CASES

- **Healthcare** — hospital FL
- **Finance** — cross-bank fraud
- **IoT / Edge** — distributed sensors
- **Research** — multi-institution
- **Government** — census / tax

SKLEARN-COMPATIBLE API 191 algorithms

- **Linear 39** — Ridge Lasso SGD
- **Ensemble 18** — RF AdaBoost
- **Clustering 14** — KMeans DBSCAN
- **Decomposition 14** — PCA NMF ICA
- **SVM 7** — SVC SVR NuSVC
- **NN 4** — MLP BernoulliRBM
- **+95** — neighbors trees NB …

PRIVACY MODES src/secretlearn/{federated_learning,secret_sharing,split_learning}/

FEDERATED LEARNING
federated_learning/
FLLinearRegression
local PYU · HEU aggregate
191 algos · 3–5×

SECRET SHARING
secret_sharing/
SSPCA
SPU MPC ABY3/CHEETAH
191 algos · max privacy

SPLIT LEARNING
split_learning/
SLMLPClassifier
model split · HEU acts
191 algos · vertical FL

INTELLIGENT SYSTEM + JAX

- **algorithm_classifier.py** — src/secretlearn/algorithm_classifier.py
- **JIT / XLA** — xlearn backend
- **jax.vmap** — batch + autodiff
- **smart fallback** — original sklearn

SECRETFLOW DEVICES

- **PYU** — plaintext party compute
- **HEU** — homomorphic aggregate
- **SPU** — MPC ABY3 / CHEETAH
- **TEE** — trusted execution

INFRASTRUCTURE & HARDWARE

- **CPU 1.5–2.5×** — local JAX
- **GPU 5–8×** — CUDA
- **TPU 9–15×** — Google
- **M1/M2 2–4×** — Metal beta

Same sklearn call surface; mode directory picks FL / SS / SL · 191 × 3 = 573 implementations

YOUTUBE-DNN-RECSYS — SYSTEM ARCHITECTURE

Covington two-tower recall · source: docs/architecture.drawio · github.com/chenxingqiang/YouTube-DNN-RecSys

DATA PROCESSING

data2tfrecords.py
src/python/data/data2tfrecords.py
random + TFRecords train/eval

read_tfrecords.py
src/python/data/read_tfrecords.py

MakeDataOne.scala
src/scala/data/MakeDataOne.scala
Spark distributed generate

MakeDataTwo.scala
src/scala/data/MakeDataTwo.scala
HDFS TFRecords

STORAGE

TFRecords

data/tfrecords/
train + eval TFRecords

data/checkpoints/
TF ckpt + SavedModel

MODEL src/python/models/dnn.py (TensorFlow Estimator)

Input
visit_items_index (5)
continuous_features (16)

Embedding
item_table-54
average pooling on history

MLP tower
dense layers → user vector u

Softmax
500-way next-item
cross entropy · Top-K

SavedModel

PREDICTION

load_dnn_model.py
src/python/models/load_dnn_model.py
SavedModel online / batch

PredictUserVector.scala
src/scala/prediction/PredictUserVector.scala

ItemEmbeddingPredictor.scala
src/scala/prediction/ItemEmbeddingPredictor.scala

ItemEmbedding.scala
src/scala/models/ItemEmbedding.scala

UTILITIES & CONFIG

tensor_board.py
src/python/utils/tensor_board.py

feature_column.py
src/python/reference/feature_column.py

self_defined_network_layer.py
src/python/reference/self_defined_network_layer.p

FeatureBuilder.scala
src/scala/models/FeatureBuilder.scala

BaseSparkLocal/Online
src/scala/core/*.scala

vars.prod/sit.properties
src/scala/config/

Python/TF path and Scala/Spark path share TFRecords + the same two-tower (history pool → user vec · item table)

RESEARCH CLI — SYSTEM ARCHITECTURE

TypeScript monorepo · packages/cli + packages/core · github.com/iechor-research/research-cli

packages/cli

- **research.tsx / TUI** — packages/cli/src
- **commands/** — mcp.ts skills.tsx hooks.tsx
- **config/config.ts** — packages/cli/src/config/config.ts
- **acp/** — acpSession.ts acpCommandHandler.ts

packages/core orchestration

- **config.ts** — packages/core/src/config/config.ts
- **research-config-manager.ts** — packages/core/src/config/
- **modelRouterService.ts** — packages/core/src/routing/
- **routingStrategy.ts** — default/classifier/composite
- **memory.ts / storage.ts** — session + compress

route

TOOLS + MCP

- **workspace tools** — fs / shell packages/core/src/tools
- **MCP** — packages/cli/src/commands/mcp.ts
- **sandbox** — config/sandbox-integration
- **packages/sdk** — programmatic embed
- **packages/a2a-server** — agent protocol
- **vscode-ide-companion** — IDE bridge

ACADEMIC MODULES packages/core/src/tools/research/

- **writing** — outline / LaTeX
- **bibliography** — arXiv / PubMed / BibTeX
- **analysis** — stats / citation net
- **investigation** — experiment codegen
- **submission** — journal matcher

PROVIDERS model-provider-config.ts + /model config

- **Gemini / Vertex**
- **OpenAI**
- **Anthropic**
- **Baidu ERNIE**
- **DeepSeek**
- **Qwen / Groq / Kimi**

CLI TUI → core config/router → provider + tools/MCP/research modules → streamed result

QUESTIONS-GEN — SYSTEM ARCHITECTURE

Qwen3-14B contest problems · src/*.py + questions_gen/ · github.com/chenxingqiang/GenQuestion

DATA

QuestionDataPreparer.py
src/QuestionDataPreparer.py

data_preparer.py
questions_gen/data/data_preparer.py

real_math_data_processor.py
src/real_math_data_processor.py

mixture
historical 50% · variants 30% · novel 20%

THREE-STAGE TRAINER

QuestionsGenTrainer.py
src/QuestionsGenTrainer.py
questions_gen/core/trainer.py
Stage1 SFT Qwen3-14B

reward

RewardCalculator.py
src/RewardCalculator.py
0.4 diff 0.3 nov 0.2 rigor 0.1 div

reward

NoveltyConstraint.py
src/NoveltyConstraint.py
GRPO 8-problem groups

DeepSeekTeacher.py
src/DeepSeekTeacher.py
Stage3 distill R1

config.py
src/config.py
questions_gen/core/config.py

VALIDATION questions_gen/validation/

model_validator.py
single-model n tests

batch_validator.py
algebra/calculus/geometry/NT

quality_evaluator.py
content clarity difficulty originality

PACKAGE / CLI / DEPLOY

main_cli.py
questions_gen/cli/main_cli.py
validate batch quality

ollama_manager.py
questions_gen/utils/ollama_manager.py

hf_utils.py
questions_gen/utils/hf_utils.py

questions_gen_training.py
scripts/questions_gen_training.py

ARTIFACTS Hugging Face xingqiang/*

stage1-fp-merged
SFT

stage2-merged-16bit
GRPO

final-merged-16bit
distill quality 0.847

Ollama questions-gen-final
local generate

Stage1 SFT → Stage2 GRPO (RewardCalculator + NoveltyConstraint) → Stage3 DeepSeekTeacher → HF/Ollama